

SHIVA ATHAPPAN

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PROFESSIONAL SUMMARY

Backend Engineer with 3+ years of experience building scalable APIs, microservices, and AI-integrated systems. Proficient in Python, FastAPI, PostgreSQL, Docker, and AWS, with production experience in caching, message queues, and CI/CD automation. Proven record of shipping features in Agile environments with measurable gains in latency, throughput, and reliability.

TECHNICAL SKILLS

- Languages:** Python, Java, SQL, JavaScript, Kotlin, C++
- Backend & APIs:** FastAPI, Flask, Node.js, RESTful APIs, Microservices, OAuth2, JWT
- Cloud & DevOps:** AWS (EC2, RDS, S3, ElastiCache), Docker, Kubernetes (K8s), GitHub Actions (CI/CD)
- Databases & Caching:** PostgreSQL, MySQL, MongoDB, SQLite, Redis
- Messaging & Streaming:** Apache Kafka, RabbitMQ, Redis (Pub/Sub, Caching, Rate Limiting)
- AI/ML:** LangChain, OpenAI API, Retrieval-Augmented Generation (RAG), FAISS, NLP, Scikit-learn, TensorFlow, Hugging Face
- Frontend:** React.js, HTML5, CSS3, Chart.js, Streamlit

EXPERIENCE

- Junior Software Developer Intern** — Connex AI, Core Engine Team Oct 2025 – Present
- Engineered Python/FastAPI backend services, reducing API latency by 20% through async optimization and query tuning.
 - Optimized 10+ SQL queries and redesigned PostgreSQL schemas, cutting average execution time by 30%.
 - Deployed 3 LangChain/OpenAI pipelines for document summarization and intent classification, boosting feature throughput by 15%.
 - Built 20+ reusable backend modules adopted across teams, trimming average build time by 25%.
- Full Stack Developer** — IdeaEdu Apr 2022 – Jun 2024
- Shipped a Python/FastAPI + SQLite learning platform with a RESTful backend supporting 40+ concurrent users.
 - Developed 8+ API endpoints with input validation and structured error handling, lowering backend response time by 40%.
 - Designed and maintained PostgreSQL schemas and integrated analytics features delivering real-time instructor insights.
 - Built React.js/Chart.js dashboards that improved student engagement by 35% and course completion by 20%.
- Freelance Math Content Developer** — Self-Employed Apr 2022 – Jun 2024
- Built 30+ Python-based math learning modules and automated grading scripts with data handling and testing practices.

PROJECTS

- LinkSnap — Production-Grade URL Shortener** 2026
- FastAPI, PostgreSQL (AWS RDS), Redis, Docker, AWS EC2/S3, GitHub Actions*
- Built a REST API with JWT authentication and rate limiting; deployed PostgreSQL on AWS RDS and containerized the app with Docker.
 - Implemented Redis caching for redirect lookups, reducing average latency by 65% versus uncached database calls.
 - Configured GitHub Actions CI/CD to AWS EC2 and logged click analytics to AWS S3 with a dedicated stats endpoint.
- AI Auto Agent — Document Q&A System** Jan 2025 – May 2025
- Python, LangChain, RAG, FAISS, FastAPI, Streamlit*
- Architected a Retrieval-Augmented Generation backend (LangChain, OpenAI API) for natural-language querying over large document corpora.
 - Indexed 10,000+ document chunks with FAISS, achieving sub-2-second query responses with 90%+ accuracy.
 - Exposed the service via FastAPI endpoints and built a 3-panel Streamlit UI (document viewer, chatbot, insights dashboard).
- CleverHire — AI Resume Screener** Jan 2025 – Apr 2025
- FastAPI, React.js, OpenAI API, PostgreSQL, GitHub Actions*
- Built an LLM-powered FastAPI screening service, reducing manual screening time by 60% with an 18% accuracy gain.
 - Designed a PostgreSQL schema storing 100+ resume results and surfaced keyword gaps through 5 Chart.js dashboards.
 - Configured a GitHub Actions CI/CD pipeline that cut deploy time by 50% and supports 5+ file formats.

EDUCATION

- Clark University** — Master of Science, Computer Science, Worcester, MA Dec 2025
- Alagappa University** — Bachelor of Science, Information Technology, Tamil Nadu, India Apr 2023

CERTIFICATIONS

- Data Science & Machine Learning** — IIIT Bangalore, India (Jun 2024)
- Data Programming and Analysis** — IIT Madras, India (Dec 2022)
- Microsoft AI Builder Challenge** — Power Platform (Aug 2021)